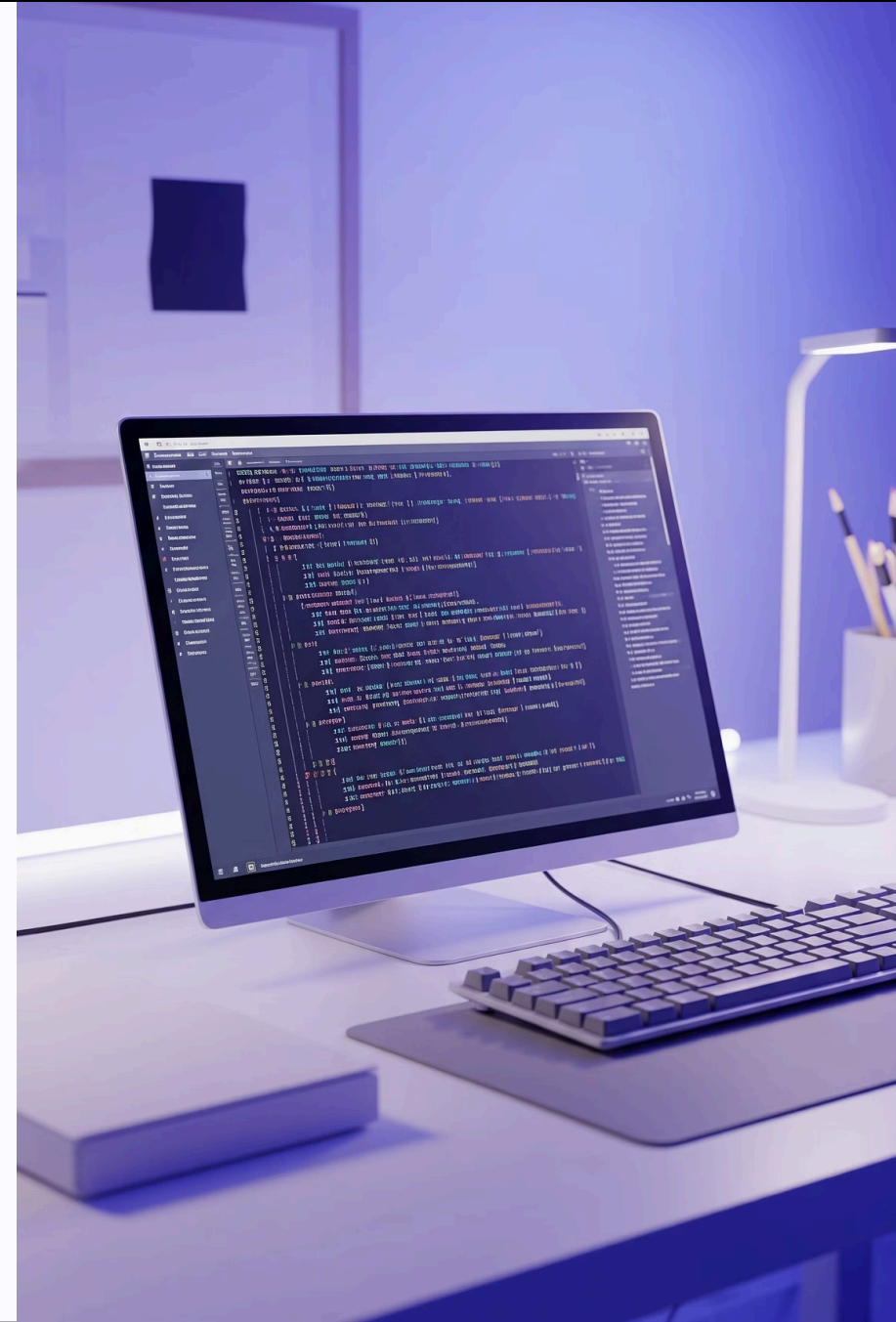


Navigating the Linux Shell: Bash Basics

A practical guide for AI engineers and developers implementing infrastructure and DevOps workflows



Why Learn Bash?



Direct OS Interface

Bash is the default shell on most Linux systems, providing direct access to operating system functions crucial for AI infrastructure tasks



Speed & Efficiency

Significantly faster than GUIs for automation and scripting - execute complex operations with minimal keystrokes



Industry Standard

Required skill for cloud computing, DevOps pipelines, and high-performance computing environments where AI systems run

Every AI engineer must be shell-fluent to work effectively in modern development environments

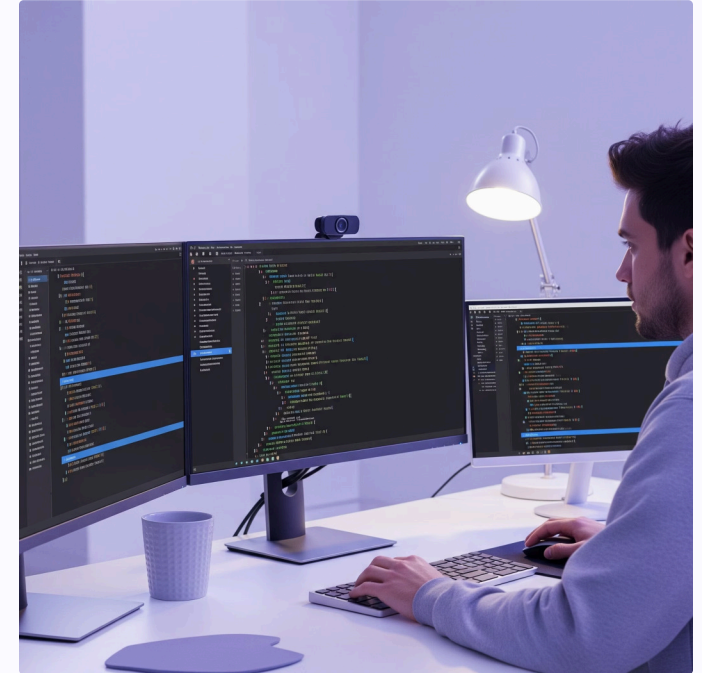
Getting Into the Shell

The shell is your direct communication channel with the operating system. You'll typically see a prompt showing:

```
username@hostname$
```

Commands are typed interactively and executed instantly when you press Enter. You can:

- Combine multiple commands with `;` or `&&`
- Access documentation with `man [command]`
- Navigate command history with up/down arrows



Access via terminal application or remote SSH connection

Essential Navigation Commands

1

pwd

Print working directory - shows your current location in the filesystem

```
$ pwd
/home/user/projects/ai-model
```

2

ls

List files and folders in the current directory

`ls -l` - detailed list with permissions

`ls -a` - show hidden files (.env, .git)

3

cd

Change directory - navigate to different locations

```
$ cd /path/to/project
$ cd .. # Move up one
directory
$ cd ~ # Go to home directory
```

File & Directory Management



Create

`touch file.txt` - create empty file

`mkdir data` - make a new directory



Copy & Move

`cp file1 file2` - copy files

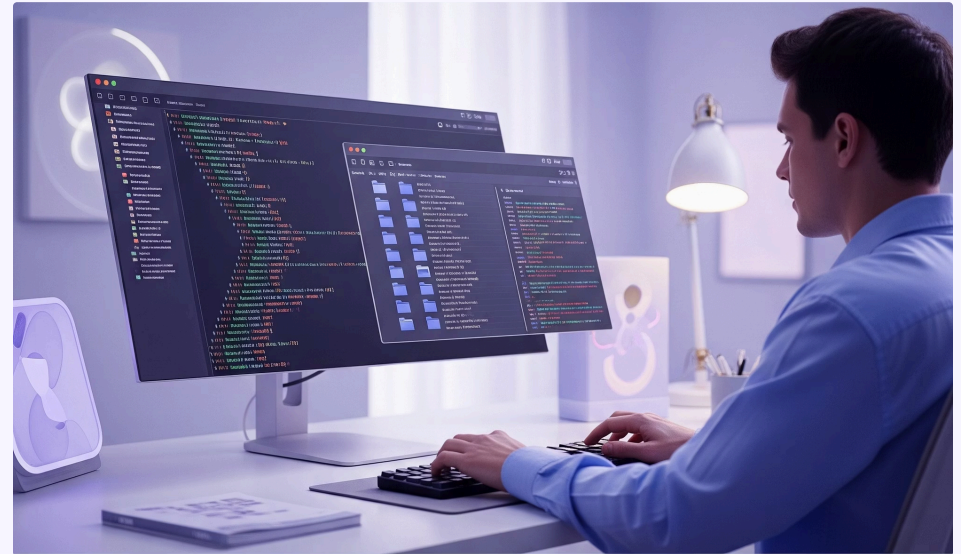
`mv file1 file2` - move or rename



Delete

`rm file.txt` - delete a file

`rm -r dir` - recursively delete dir



Efficient file management is crucial when working with large datasets and model files

Viewing & Editing Files

Quick View Commands

- `cat file.txt` - print entire file contents
- `less file.txt` - scrollable file viewer
- `head -n 10 file.txt` - first 10 lines
- `tail -n 10 file.txt` - last 10 lines

Text Editors

- `nano` - Simple, beginner-friendly editor
- `vim` - Powerful, steep learning curve
- `code` - VS Code, if installed

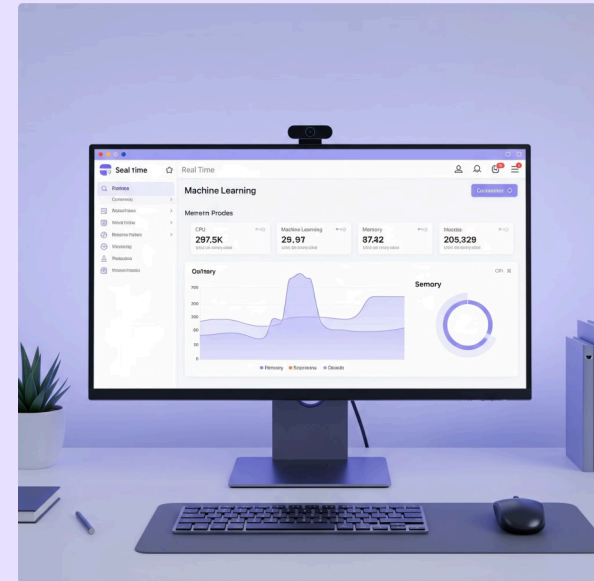
Choose based on task complexity and your skill level

When inspecting AI logs or configuration files, these commands help you quickly identify issues without loading full editors

Working with Processes

- `ps aux` - List all running processes with details
- `top` or `htop` - Real-time process monitor
- `kill [PID]` - Terminate process by ID
- `jobs` - List background tasks
- `fg/bg` - Move jobs to foreground/background

⚠ Critical for AI Workflows: Knowing how to monitor and terminate processes is essential for stopping runaway training jobs that consume excessive resources



Process management is crucial when running resource-intensive AI workloads

Redirection & Pipes

Output Redirection



> - Redirect output to file (overwrites)

>> - Append output to file

```
$ python train.py > training_log.txt
```

Input Redirection



< - Read input from file

```
$ sort < unsorted_data.txt > sorted_data.txt
```

Pipes



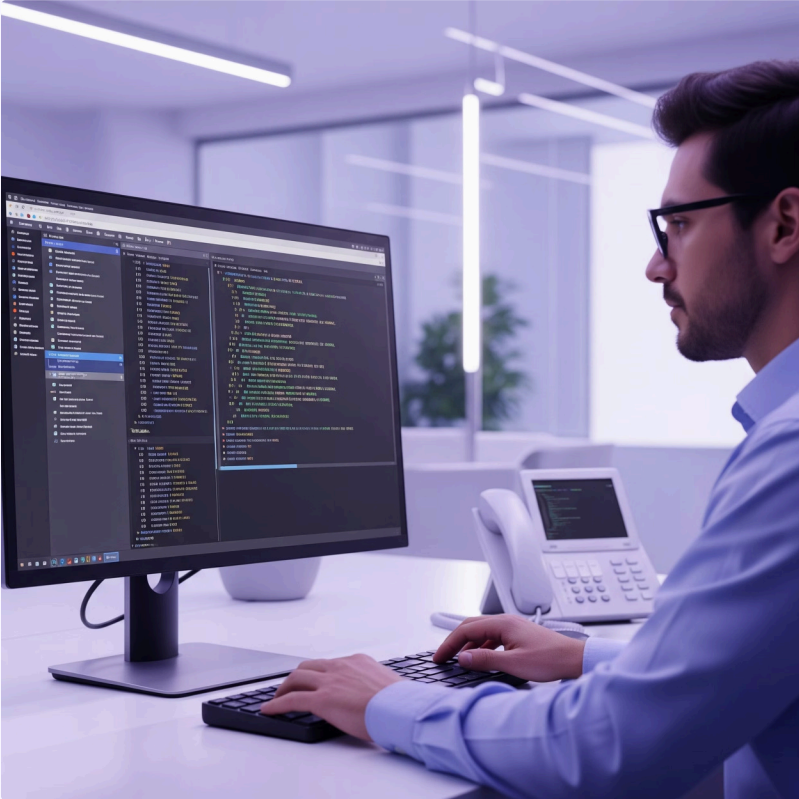
| - Send output of one command as input to another

```
$ cat data.txt | grep "error" > errors.txt
```

These operators are the building blocks for creating powerful data processing pipelines



Why Bash Matters for AI Engineers



Beyond Basic Commands

- Speeds up dataset preparation & automation workflows
- Core skill for working with Docker/Kubernetes environments
- Essential component in MLOps pipelines and CI/CD scripts
- Universal across all Linux distributions where AI systems run
- Builds foundation for advanced scripting and custom tool development

Mastering Bash is a multiplier for your effectiveness as an AI engineer - it's the glue that connects different components of your AI infrastructure